

Project Report: Multimodal Fashion & Context Retrieval Search Engine

This report outlines the design, implementation, evaluation, and future roadmap of a specialized search engine built for zero-shot fashion and environmental retrieval. The system is designed to overcome the core limitations of standard vision-language models--specifically compositionality (distinguishing item-color associations) and fine-grained fashion attribute matching.

1. Architectural Exploration & Tradeoffs

To solve zero-shot natural language image retrieval for fashion, we analyzed three primary approaches:

Approach	Architecture Description	Pros	Cons	When to Use
1. Vanilla CLIP (Baseline)	Encode the full image and full text query using standard CLIP (e.g. ViT-B/32) and compute cosine similarity.	- Extremely fast to index and query. - Excellent zero-shot general context.	- Struggles with compositionality (e.g. confuses "red shirt, blue pants" with "blue shirt, red pants"). - Misses small, fine-grained details (e.g. ties, accessories).	Baseline, general search, high-throughput systems.
2. VLM-based Image Captioning (Llava/Moondream)	Run a Vision-Language Model on all images to generate detailed text captions. Index the captions with a text search engine (BM25 or Dense Text Retriever).	- Highly detailed context and relationships captured in text.	- Extremely slow indexing on CPU (~3-5s per image). - Captions depend heavily on VLM prompt quality.	Small datasets, high-compute environments, query-heavy text search.
3. Multi-Vector Compositional Search (MVCS) - (Chosen)	Run a lightweight person detector (YOLOS-Tiny) to crop upper/lower garments. Index three normalized vectors per image: Global, Upper, and Lower. Decompose queries and score compositely.	- Solves compositionality mathematically. - Fast inference on CPU (~0.8s total). - Captures fine-grained color/style bounds.	- Slightly larger database size (3 vectors instead of 1). - Requires basic anatomical heuristics.	Fashion-specific retrieval, high precision, compositional queries.

2. Chosen Approach: Multi-Vector Compositional Search (MVCS)

Our implemented search engine uses a Multi-Vector Compositional Search (MVCS) architecture. It breaks down the retrieval task into a multi-channel alignment problem:

```
????????????????????
?  Natural Language   ?
?      Query          ?
????????????????????
?
[ Syntactic Parser ]
```

```

????????????????????????????????????????????????????????
?                                     ?                                     ?
[ Global Query ]   [ Upper Query ]   [ Lower Query ]
?                 ?                 ?
(CLIP Text)        (CLIP Text)        (CLIP Text)
?                 ?                 ?
Vector Global      Vector Upper        Vector Lower
?                 ?                 ?
????????????????????????????????????????????????????????
?
[ Composite Cosine Similarity ]
?
????????????????????????????????????????????????????????
?                 ?                 ?
Vector Global      Vector Upper        Vector Lower
(Full Image)       (Upper Crop)         (Lower Crop)
?                 ?                 ?
?                 ?                 ?
(CLIP Image)       (CLIP Image)         (CLIP Image)
?                 ?                 ?
?                 ?                 ?
?                 [ Cropping ]
?                 ?
?                 (YOLOS-Tiny)
????????????????????????????????????????????????????????
?
????????????????????
? Raw Image ?
????????????????????

```

Key Technical Elements:

1. Person Detection & Crop Heuristics (Part A - Indexer):

- Uses `hustvl/yolos-tiny` to detect people.
- For the main person, it crops the Upper Body (y-coordinates `ymin + 0.10\*h` to `ymin + 0.65\*h`) and Lower Body (y-coordinates `ymin + 0.50\*h` to `ymax`).
- If no person is found (e.g. flat lays), it defaults to top-half and bottom-half image crops.

2. Normalized Vector Database (Part A - Indexer):

- Extracts 512-dimensional normalized embeddings for the full image and crops using `openai/clip-vit-base-patch32`.
- Stores these vectors in a modular SQLite database (`fashion\_images` table), ensuring scalability and persistence.

3. Query Parser (Part B - Retriever):

- Syntactically parses queries into `global`, `upper`, and `lower` components by segmenting around prepositions ("with", "sitting on", "and", "in a").

4. Dynamic Composite Scoring (Part B - Retriever):

- Evaluates search results using a weighted combination of cosine similarities:

$$\text{Score} = w_{\text{global}} \cdot \text{Sim}(E_{\text{text, global}}, E_{\text{img, global}}) + w_{\text{upper}} \cdot \text{Sim}(E_{\text{text, upper}}, E_{\text{img, upper}}) + w_{\text{lower}} \cdot \text{Sim}(E_{\text{text, lower}}, E_{\text{img, lower}})$$

- Weights are dynamically set:
 - Both upper & lower specified: $w_{\text{global}} = 0.40$, $w_{\text{upper}} = 0.30$, $w_{\text{lower}} = 0.30$.
 - Only upper specified: $w_{\text{global}} = 0.50$, $w_{\text{upper}} = 0.50$, $w_{\text{lower}} = 0.00$.
 - Only context specified: $w_{\text{global}} = 1.00$, $w_{\text{upper}} = 0.00$, $w_{\text{lower}} = 0.00$.
-

3. Evaluation & Verification Results

The search engine was evaluated on the 5 benchmark prompts specified in the assignment. The queries are decomposed and searched against the indexed database of images.

Benchmark Query Decomposition:

1. Attribute Specific: *"A person in a bright yellow raincoat."*
 - Upper Query: "a bright yellow raincoat" | Lower Query: "yellow raincoat or pants"
 - Successfully targets the bright yellow raincoat crop and coordinates global similarity.
2. Contextual/Place: *"Professional business attire inside a modern office."*
 - Upper Query: "professional business blazer or shirt" | Lower Query: "professional business trousers or skirt"
 - Successfully matches blazer textures in the upper crop and office-like environment embeddings globally.
3. Complex Semantic: *"Someone wearing a blue shirt sitting on a park bench."*
 - Upper Query: "a blue shirt" | Lower Query: "pants or jeans"
 - Correlates the blue shirt crop with the bench/park global background context.
4. Style Inference: *"Casual weekend outfit for a city walk."*
 - Upper Query: "casual t-shirt or hoodie or top" | Lower Query: "jeans or casual pants or skirt"
 - Evaluates style vibe using the query expansion heuristics.
5. Compositional: *"A red tie and a white shirt in a formal setting."*
 - Upper Query: "a red tie and a white shirt" | Lower Query: "formal suit trousers or skirt"
 - Isolates the upper crop to prevent matching false-positive combinations (e.g. white tie and red shirt).

Note:

Note: The automated script generates annotated top-5 search grids (with scores and bounding boxes) in the artifacts directory. These grids show clear visual alignment with the prompts.

4. Future Extensions

A. Location & Weather Integration

To extend search to encompass location context (e.g. "rainy London street", "sunny Miami beach") and weather dynamics:

1. Dual-Model Metadata Tagging:

- Integrate a scene classifier (like Places365) or a general image tagger to extract semantic environment tags (e.g., `snowing`, `raining`, `sunny`, `beach`, `urban`).
- Store these metadata tags as indexing attributes in SQLite.

2. External Weather/Geotag Databases:

- If images contain metadata (EXIF headers for GPS coordinates and timestamps), integrate a weather archive API (like Open-Meteo) to dynamically assign weather context (e.g. temperature, precipitation) during indexing.

3. Query Expansion & Semantic Reranking:

- Expand queries like "London street" at run-time (using a lightweight dictionary or LLM) to include terms like "rainy", "wet pavement", "city buildings", "gloomy weather" and weight the global context vector accordingly.

B. Strategies to Improve Precision

1. Fine-Tuning CLIP (Contrastive Adaptation):

- Fine-tune CLIP using a contrastive loss on fashion datasets (like DeepFashion2 or Fashionpedia) to align fashion-specific text descriptors (e.g., "houndstooth pattern", "A-line skirt") with fashion crops.

2. Visual Question Answering (VQA) Reranking:

- Implement a fast VQA model (like Moondream or BLIP) to act as a Stage-2 reranker.
- Search the database using MVCS to get the top 50 matches, then ask the VQA model: `"Is this person wearing a bright yellow raincoat? Answer yes or no."` Filter/rerank results based on VQA confidence.

3. Text Query Expansion using LLMs:

- Use an LLM to generate descriptive variations of the search query (e.g., expanding "business attire" to "button-down shirt, suit jacket, tie, trousers") before encoding.

5. Codebase Link & Structure

- GitHub Repository Link: <https://github.com/Aragog540/Multimodal-Fashion-Context-Retrieval>

The codebase is highly modular, separating feature extraction, database storage, and retrieval logic:

```
glance assgn/  
?  
??? indexer/  
?   ??? detector.py      # YOLO5-Tiny object detection and crop generation  
?   ??? embedder.py      # CLIP embedding extraction (L2-normalized)  
?   ??? pipeline.py      # Batched indexing pipeline  
?  
??? retriever/  
?   ??? parser.py        # Rule-based query decomposer  
?   ??? search.py        # Search scoring logic (Dynamic weight composite  
matching)  
?   ??? cli.py           # Search command CLI  
?  
??? utils/  
?   ??? database.py      # SQLite vector database connector  
?  
??? app.py               # Streamlit Web UI Dashboard  
??? test_queries.py      # Automated benchmark query evaluator and grid generator
```

```
??? run.py # Command-line entrypoint script
```

Usage Instructions:

- To run the Indexer (Part A):

```
python run.py index --limit 1000 --batch 16
```

- To launch the Interactive Web UI:

```
streamlit run app.py
```

- To run the Retriever Search (Part B via CLI):

```
python run.py search "Someone wearing a blue shirt sitting on a park bench"  
--show
```

- To run the full evaluation suite and generate visual artifacts:

```
python test_queries.py
```